

19-202-0703 CRYPTOGRAPHY AND NETWORK SECURITY

Course Outcomes:

On completion of this course the student will be able to:

1. *Identify security issues in the network and provide data security over the network.*
2. *Familiarise cryptographic algorithms, hash codes and digital signatures.*
3. *Examine the issues and structure of authentication service and electronic mail security.*
4. *Familiarise with network security protocols used to protect against threats in the network.*
5. *Familiarise methods for authentication, access control, intrusion detection and prevention.*

Module I

Introduction- Security problem in computing, Security in networks. Elementary cryptography- Introduction-Substitution, Transposition, Hill Ciphers and Affine Ciphers. Review of number Theory- Modular arithmetic.

Module II

Encryption Algorithms- Symmetric key encryption - DES- The Feistel cipher structure. The avalanche effect. Modes of operations of DES algorithm. AES-S-boxes.

Module III

Public Key Encryption- RSA Cryptosystem. Primality testing-Miller-Rabin Algorithm. Key Management-Diffie- Hellman- Cryptosystem-ElGamal Scheme Elliptical Curve Cryptography and Elliptic curves over Finite Fields. Hash Algorithms- SHA 1 and SHA-512 MAC-HMAC.

Module IV

Authentication protocols, Authentication applications-Kerberos, X.509, Digital Signature, Secure e-mail, SSL, IP Security - Intruders -Malicious Software, Firewalls.

References:

1. Cryptography and Network Security – Principles and Practice: William Stallings, Pearson Education, 6th Edition.
2. Cryptography and Network Security: Atul Kahate, Mc Graw Hill, 3rd Edition.
3. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1st Edition.
4. Cryptography and Network Security : Forouzan Mukhopadhyay, Mc Graw Hill, 3rd Edition.
5. Introduction to Network Security: Neal Krawetz, CENGAGE Learning.
6. Principles of Computer Security: WM. Arthur Conklin, Greg White, TMH.